You believe in conservation of matter. No object can ever be created from nothing, yet this object was. The only explanation you can find is shown in the right half of Figure 22.2. A three-dimensional bulk being —a sphere—passed through your brane. As it passed through, you saw in your brane its changing two-dimensional cross section. The cross section began with a point at the sphere's south pole (top right). It expanded to a maximal circle, the sphere's equatorial plane (middle right). It then shrank to a point, the sphere's north pole, and disappeared (bottom right).

Imagine what would happen if a 3D human being, living in the 3D bulk, passed through your 2D brane. What would you see?

Bulk Beings from the Fifth Dimension, Passing Through Our 3D Brane

Suppose that our universe, with its three space and one time dimensions, really does live in a five-dimensional bulk (four space and one time). And suppose there are "hyperspherical beings" who live in the bulk. Such a being would have a center and a surface. Its surface would consist of all points, in four space dimensions, that are some fixed distance from the center, for example, 30 centimeters. The bulk being's surface would have three dimensions and its interior would have four.

Suppose that this hyperspherical bulk being, traveling in the bulk's out direction or back direction, were to pass through our brane. What would we see? The obvious guess is correct. We would see spherical cross sections of the hypersphere (Figure 22.3).